

OS Goals & Selection

1. Goals of Operating System

The primary goals of an operating system (OS) are to provide an easy to use and convenient environment for executing user programs.

Goal	Description
User Convenience	It should be easy to use, providing a user-friendly interface and making it simple to interact with the system.
Program Execution	It facilitates the execution of user programs, providing the necessary environment and services for them to run.
Resource Management	The OS manages and allocates the computer's resources, including the CPU, memory, disk storage and input/output devices, to ensure fair utilization.
Security	The OS protects the system and user data from unauthorized access, ensuring confidentiality, integrity and availability of information.
Efficient Resource Utilization	It should aim to maximize the performance and utilization of computer resources like CPU, Memory and I/O devices, ensuring that the system runs smoothly and efficiently.
Reliability	It should be robust and reliable, able to handle errors and exceptions gracefully, ensuring that the system continues to operate smoothly. It should be modular in design and easy to debug.

2. How to Choose the Operating System?

There are so many factors to be considered while choosing the best Operating System for our use. These factors are mentioned below.

Factor	Description
Price Factor	Price is one factor to consider when choosing an operating system, as some OSs are free, like Linux, but there are some more OS that are paid like Windows and macOS.
Accessibility Factor	Some Operating Systems are easy to use like macOS and iOS, but some OS are a little bit complex to understand like Linux. So, you must choose the Operating System in which you are more accessible.
Compatibility factor	Some Operating Systems support few applications whereas some Operating Systems support more applications. You must choose the OS which supports the applications required by you.
Security Factor	Security is an important factor when choosing the correct operating system. Both macOS and Microsoft Windows provide strong security features, and the level of security mainly depends on proper configuration, regular updates, and how the system is used.

3. 32-bit vs 64-bit

The supplied notes contain the heading "32 vs 64 bit" but do not provide further explanation under this topic. No additional material has been added here.
